

Online Food Delivery Analysis

Project Overview

The Online Food Delivery Analysis project is designed to analyze food delivery order data and understand customer, restaurant, delivery, financial, and operational patterns.

The project uses a food delivery dataset containing information about customers, restaurants, cuisines, orders, delivery times, distances, order values, discounts, payment methods, cancellations, ratings, and profit margins.

The main goal of this project is to understand food delivery business performance and identify useful trends related to orders, revenue, profit, delivery performance, cancellations, customer behavior, and restaurant performance using Python, Pandas, MySQL, SQL, and Streamlit.

The project also provides an interactive dashboard where important business metrics and findings can be viewed easily.

Project Objectives

The main objectives of this project are:

- Analyze food delivery order data.
- Understand customer and order patterns.
- Clean and prepare the raw dataset for analysis.
- Handle missing and invalid data appropriately.
- Analyze order volume and monthly order trends.
- Analyze revenue, discounts, and final order amounts.
- Analyze profit and profit margins.
- Study delivery time and delivery distance.
- Analyze order cancellation patterns and cancellation reasons.
- Analyze delivery and restaurant ratings.
- Understand payment mode usage.
- Analyze orders by city, area, and cuisine.
- Store the cleaned data in MySQL.
- Perform SQL-based business analysis.
- Create a Streamlit dashboard to present important findings.

Data Source

The dataset contains information on **100,000 food delivery orders**. It includes the following:

- Customer and Restaurant details
- Cuisine Type
- Order date and time
- Delivery time and distance
- Order value, Discounts, Final amount and Profit margin
- Payment method
- Order status
- Cancellation reason
- Delivery ratings and Restaurant ratings
- Peak-hour information

The dataset contains **25 original columns**.

Features in the Dataset

The dataset contains customer, restaurant, order, delivery, financial, and rating-related features. Additional derived features were also created during the data preparation stage.

Feature	Datatype	What it means	Why needed
Order_ID	string	Unique identifier assigned to each order	Used to identify individual orders
Customer_ID	string	Identifier of the customer	Used for customer-level analysis
Customer_Age	numeric	Age of the customer	Useful for customer segmentation
Customer_Gender	string	Gender of the customer	Used to analyze customer demographics
City	string	City associated with the order	Used for geographic analysis
Area	string	Area associated with the order	Used for detailed geographic analysis
Restaurant_ID	string	Unique restaurant identifier	Used to identify restaurants
Restaurant_Name	string	Name of the restaurant	Used for restaurant-level analysis
Cuisine_Type	string	Type of cuisine ordered	Used to understand cuisine preferences
Order_Date	date	Date on which the order was placed	Used for time-based analysis
Order_Time	time	Time at which the order was placed	Used to study ordering patterns
Delivery_Time_Min	numeric	Delivery time in minutes	Used to analyze delivery performance
Distance_km	numeric	Delivery distance in kilometers	Used to study delivery distance
Order_Value	numeric	Original value of the order	Used for financial analysis

Discount_Applied	numeric	Discount applied to the order	Used to analyze discount behavior
Final_Amount	numeric	Final order amount after discount	Used to calculate revenue
Payment_Mode	string	Payment method used by customer	Used to understand payment preferences
Order_Status	string	Status of the order	Used to analyze delivered and cancelled orders
Cancellation_Reason	string	Reason for order cancellation	Used to identify cancellation patterns
Delivery_Partner_ID	string	Identifier of delivery partner	Used for delivery-related analysis
Delivery_Rating	numeric	Customer rating for delivery	Used to measure delivery satisfaction
Restaurant_Rating	numeric	Customer rating for restaurant	Used to measure restaurant satisfaction
Order_Day	string	Day information associated with the order	Used for day-wise analysis
Peak_Hour	boolean	Indicates whether the order occurred during peak hours	Used to analyze peak demand
Profit_Margin	numeric	Profit margin associated with the order	Used for profitability analysis

Data Collection

The dataset was provided as a CSV file and loaded into Python using Pandas.

The data collection/loading process included:

1. Loading the CSV file into a Pandas DataFrame.
2. Checking the number of rows and columns.
3. Inspecting column names.
4. Checking data types, duplicate records, and missing values.
5. Examining unique and categorical values.
6. Performing initial statistical analysis.

Data Cleaning

After loading the dataset, several data-cleaning steps were performed. The purpose of data cleaning was to improve data quality while preserving useful business information.

Handling Duplicate Records

Duplicate records were checked using the complete dataset.

Duplicate Order_ID values were also checked.

The analysis found:

- Duplicate rows: **0**
- Duplicate Order IDs: **0**

Therefore, no records were removed because of duplication.

Handling Missing Values

Missing values were checked for all columns. Several columns contained missing values, including:

- Customer_Age
- Customer_Gender
- City
- Area
- Cuisine_Type
- Order_Date
- Order_Time
- Delivery_Time_Min
- Distance_km
- Order_Value
- Discount_Applied
- Final_Amount
- Payment_Mode
- Delivery_Rating
- Cancellation_Reason

The missing values were not handled in the same way for every column. The treatment depended on the meaning of the missing value.

Handling Numerical Missing Values

The numerical features of **Customer_Age, Delivery_Time, Distance_km and Order_Value** did not contain significant outliers. Therefore, missing values in these columns were handled using the **median** of the respective feature.

Handling Categorical Missing Values

Missing values in the following categorical columns were handled together: **Customer_Gender, City, Area, Cuisine_Type, Payment_Mode**

These fields were handled together during the cleaning process. Instead of assuming a missing city, area, cuisine, or gender, missing categorical values were treated as an **Unknown** category.

This approach preserves the original orders while making the missing information explicit.

For example: NaN → Unknown

Handling Cancellation Reason

Cancellation_Reason requires special handling because its meaning depends on Order_Status.

- For delivered orders, a cancellation reason does not apply. Therefore, missing cancellation reasons for delivered orders were represented as: **Not Cancelled**
- For cancelled orders where the reason was missing, the value was represented as: **Unknown**

This approach keeps the difference between a delivered order, a cancelled order with a known reason, and a cancelled order where the reason is unavailable.

Restaurant Rating Validation

Restaurant ratings were checked to make sure that the values followed the expected **1–5 rating scale**. The analysis found: 18,019 records with Restaurant_Rating greater than 5. These values were considered invalid because the expected maximum rating is 5.

The values were therefore capped at the upper valid limit of **5**. This prevents invalid ratings from affecting average restaurant-rating calculations.

Peak Hour Handling

This one is different from the other categorical columns. Since Peak_Hour is a Boolean field, we shouldn't fill out the missing values with "Unknown" because that would change its data type.

Also, we shouldn't automatically use the mode (True or False) because that would invent peak-hour information for almost 33,000 orders.

So, we use a third category such as Unknown but first convert the column to a categorical/text representation.

Delivery Rating Handling

- **Cancelled orders:** Delivery ratings were set to **NaN** because a rating is not applicable when an order is cancelled.
- **Delivered orders:** The original delivery ratings were retained.
- **Missing ratings for delivered orders:** Missing delivery ratings were filled using the **median rating of the delivered orders only**. This ensures that cancelled orders do not influence the imputation.

Handling Discount Applied

Missing values in Discount Applied were handled using the **median value of 50**, as the discount data contained potential extreme values.

The IQR method was then used to identify possible outliers in the discount values. A total of **16,600 outliers** were identified based on the IQR boundaries.

These values were retained because they may represent valid higher discounts rather than errors.

Handling Final Order Amount

The **Final_Order_Amount** should normally be **0 or greater**, as an order amount cannot be negative.

During data validation, records were identified where **Discount_Applied > Order_Value**. These records also contained **negative Final_Order_Amount values**, indicating potentially invalid discount information.

Instead of deleting these records, the invalid values were **corrected using appropriate values** to ensure that the Final_Order_Amount remained **0 or greater**.

Removing Order Time

The **Order_Time** column was removed from the dataset because all values in the column were recorded as **0.00**. Since the column contained no meaningful variation or useful information for analysis, it was dropped from the dataset.

Handling Order Date and Delivery Rating Missing Values

- **Order_Date** missing values were kept as NaN because an actual date cannot be safely inferred without affecting time-based analysis.
- **Delivery_Rating** missing values were also kept as NaN because a rating should only represent an actual customer-provided delivery rating.
- This prevents incorrect dates or ratings from being introduced into the analysis.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to identify patterns and trends in food delivery data.

The analysis included the distribution of **Order Value and Delivery Time**, along with **city-wise and cuisine-wise order analysis**.

Weekday and weekend demand, distance and delivery-time relationships, and cancellation reasons were also analyzed.

Correlation analysis was performed among important numerical features to understand their relationships and identify useful patterns.

Feature Engineering

Additional features were created from the existing dataset to make analysis easier.

The derived features included:

- Profit Margin Percentage
- Customer Age Group
- Delivery Performance Categories
- Day of Week

Database

After cleaning and transforming the data, the dataset was stored in a MySQL database.

Python was connected to MySQL using SQLAlchemy. The cleaned order-level data was loaded into the database and used for SQL-based analysis. The database provides a structured environment for performing business queries on the cleaned dataset.

The main table contains the original order information together with the derived analytical features required for the project.

Analyst Tasks (EDA & Analytics)

EDA was performed to understand customer behavior, revenue and profitability, delivery performance, restaurant performance, and operational patterns in the food delivery dataset.

Customer & Order Analysis

- Identified **top-spending customers** based on total order value.
- Analyzed **age groups vs. order value** to understand customer spending patterns.
- Compare **weekday vs. weekend orders** to identify ordering trends.

Revenue & Profit Analysis

- Analyze **monthly revenue and profit trends**.
- Studied the **impact of discounts on profitability**.
- Identified **high-revenue cities and cuisines**.

Delivery Performance

- Compare **average delivery time across cities**.

- Analyzed the relationship between **distance and delivery time**.
- Studied **delivery ratings vs. delivery time**.

Restaurant Performance

- Identified **top-rated restaurants**.
- Analyzed **cancellation rates by restaurant**.
- Compared **cuisine-wise order and revenue performance**.

Operational Insights

- Analyzed **peak-hour demand**.
- Identified **customer payment mode preferences**.
- Analyzed **cancellation reasons** to identify major causes of cancelled orders.

These analyses helped identify important **customer, revenue, delivery, restaurant, and operational insights**, which were later presented through the Streamlit dashboard.

Summary of EDA & Analytics

The EDA phase provided a comprehensive view of the food delivery business from multiple perspectives:

Analysis Area	Key Business Question
Customer Analysis	Who are the highest-value customers?
Age Group Analysis	How does spending vary by customer age?
Order Pattern Analysis	When do customers place more orders?
Revenue Analysis	How does revenue change over time?
Discount Analysis	How do discounts affect profitability?
City & Cuisine Analysis	Which markets and cuisines generate higher revenue?
Delivery Analysis	What factors affect delivery performance?
Restaurant Analysis	Which restaurants perform well and which have higher cancellations?
Peak Hour Analysis	When is demand highest?
Payment Analysis	Which payment methods do customers prefer?
Cancellation Analysis	Why are orders being cancelled?

Streamlit Dashboard

A Streamlit dashboard was created to present the analysis in a simple and interactive way. The dashboard allows users to understand the major business metrics without directly running Python code or SQL queries.

Dashboard Summary Metrics

The main summary metrics include:

- **Total Orders**
- **Total Revenue**
- **Average Order Value**
- **Average Delivery Time**
- **Cancellation Rate**
- **Average Delivery Rating**
- **Profit Margin %**

These KPIs provide a quick overview of the overall business performance.

Dashboard Filters

Interactive filters were included to allow users to analyze specific portions of the dataset.

The dashboard supports filtering based on dimensions such as:

- City
- Cuisine
- Date Range
- Order Status

The KPI values and visualizations respond to the selected filters. This allows users to perform more focused analysis.

For example, users can investigate:

- Performance of a particular city
- Performance of a particular cuisine
- Orders during a particular period

- Delivered vs cancelled orders

Dashboard Visualizations

The dashboard includes visualizations for important business trends.

Monthly Order Trend

Shows the number of orders recorded during each month.

Monthly Revenue and Profit

A multi-series bar chart compares monthly:

- Revenue
- Profit

This makes it easier to understand financial performance over time.

Cancellation Analysis

A donut/pie chart shows the distribution of cancellation reasons.

Geographic Analysis

City and area information can be used to analyze geographic order performance.

Monthly Orders by Day of Week

The **Monthly Orders by Day of Week** heatmap shows the distribution of food delivery orders across different months and days of the week.

Technologies Used

The following technologies were used in the project:

Technology	Usage
Python	Data loading, cleaning, transformation, and analysis
Pandas	Data manipulation and preprocessing
NumPy	Numerical operations
Matplotlib	Exploratory data visualization
Seaborn	Statistical visualization during EDA
Plotly	Interactive dashboard visualizations

MySQL	Storage of cleaned order data
SQL	Business analysis and querying
SQLAlchemy	Connecting Python with MySQL
Streamlit	Creating the interactive dashboard
VS Code	Development environment

Future Analysis and Enhancements

The current project provides useful insights into food delivery operations and business performance. The project can be further improved with the following enhancements:

Customer Segmentation

Customer behavior can be analyzed using:

- Age
- Order frequency
- Order value
- Cuisine preference
- Spending patterns

Customer segmentation techniques can then be used to identify different customer groups.

Restaurant Performance Analysis

Restaurants can be compared using:

- Order volume
- Revenue
- Ratings
- Cancellation rate
- Profitability

Predictive Analysis

Machine learning can be applied to predict:

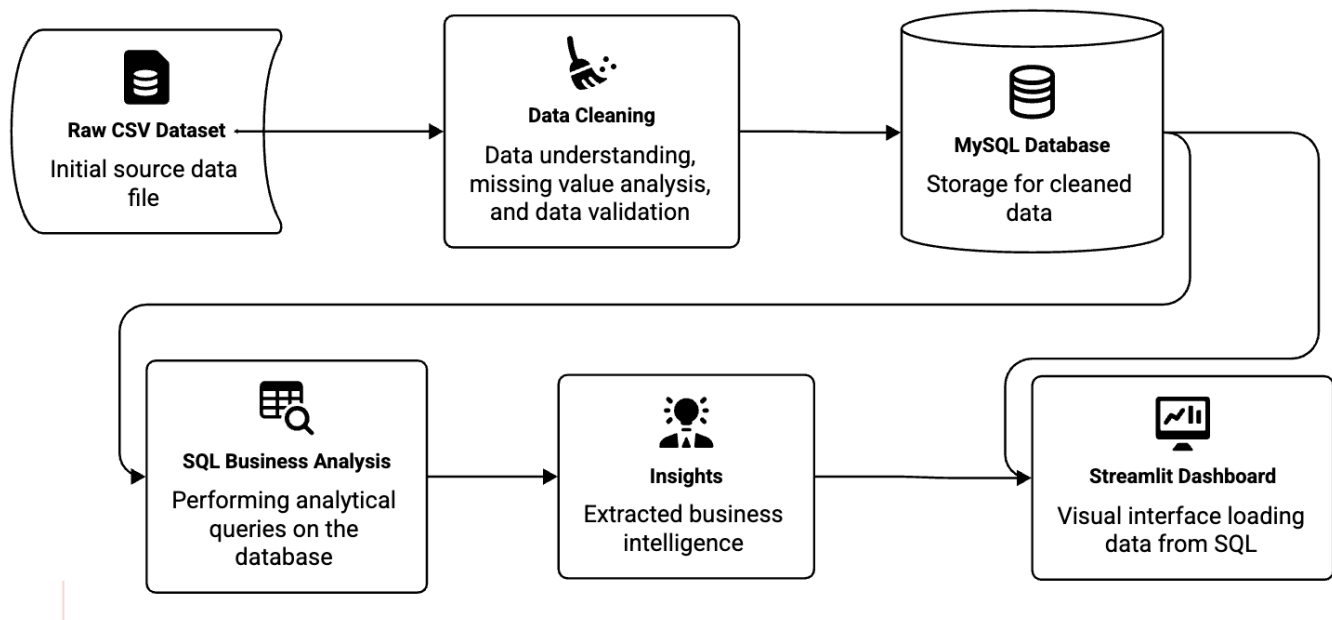
- Order cancellations
- Delivery time
- Customer spending
- Customer repeat behavior

Demand Forecasting

Historical monthly and daily order patterns can be used to forecast future food delivery demand.

Project Workflow

The overall workflow of the project is:



Conclusion

The Online Food Delivery Analysis project provides a detailed analysis of **100,000 food delivery orders** and their related customer, restaurant, delivery, and financial information.

The data was carefully cleaned and validated to handle missing values, duplicates, inconsistent ratings, outliers, and invalid financial records. Additional features were also created to support meaningful analysis.

The analysis provides insights into **order trends, cancellations, revenue, profit, discounts, delivery performance, ratings, payment methods, cuisine preferences, and geographic patterns**.

The final results help identify important patterns and trends in food delivery operations and provide a clear understanding of customer behavior, business performance, and operational factors.

Overall, the project demonstrates how raw food delivery data can be transformed into **meaningful insights that support better business understanding and decision-making**.